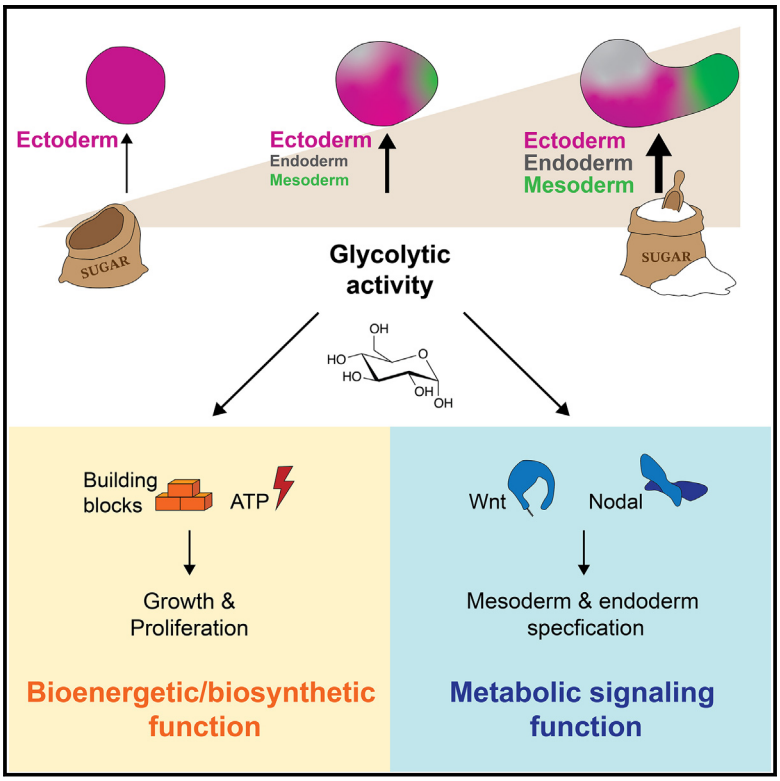
Cell Stem Cell

Article

Glycolytic activity instructs germ layer proportions through regulation of Nodal and Wnt signaling

Graphical abstract



Highlights

- Glycolysis is needed for mesoderm and endoderm formation in gastruloids
- Exogenous glucose levels have a dose-dependent effect on germ layer proportions
- Nodal, Wnt, and Fgf signaling pathway activity is dependent on glycolysis
- Glycolysis-inhibited gastruloids can be rescued by Nodal or Wnt signaling activation

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In brief

Stapornwongkul and colleagues investigate the role of metabolism during germ layer specification using a stem cell-based model system for gastrulation. They demonstrate that beyond its bioenergetic function, glycolysis also acts as an upstream regulator of important developmental signaling pathways, thereby controlling mesoderm and endoderm induction in a dose-dependent manner.

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Glycolytic activity instructs germ layer proportions through regulation of Nodal and Wnt signaling

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SUMMARY

Metabolic pathways can influence cell fate decisions, yet their regulative role during embryonic development remains poorly understood. Here, we demonstrate an instructive role of glycolytic activity in regulating signaling pathways involved in mesoderm and endoderm specification. Using a mouse embryonic stem cell (mESC)-based *in vitro* model for gastrulation, we found that glycolysis inhibition increases ectodermal cell fates at the expense of mesodermal and endodermal lineages. We demonstrate that this relationship is dose dependent, enabling metabolic control of germ layer proportions through exogenous glucose levels. We further show that glycolysis acts as an upstream regulator of Nodal and Wnt signaling and that its influence on cell fate specification can be decoupled from its effects on growth. Finally, we confirm the generality of our findings using a human gastrulation model. Our work underscores the dependence of signaling pathways on metabolic conditions and provides mechanistic insight into the nutritional regulation of cell fate decision-making.

INTRODUCTION

Research in the field of stem cell biology has been critical in uncovering the mechanisms that underlie the intricate interplay between metabolism and cell fate determination.[1–3] It is now widely acknowledged that metabolic pathways not only fulfill the bioenergetic needs of cells but also act as regulators of differentiation. The underlying mechanisms range from metabolite-driven post-translational modifications and metabolite-protein interactions to moonlighting metabolic enzymes and can affect the epigenetic as well as the signaling state of cells.[4–7] This perspective has gained support from *in vivo* studies that underscore the mechanistic role of metabolism during embryonic development.[8–17] Regulation of cellular metabolic state has been exploited to enhance the efficiency of differentiation and reprogramming protocols.[18] Therefore, a comprehensive understanding of the complex interactions between metabolism, signaling, and differentiation will open new avenues for engineering reproducible tissue patterns *in vitro*. In a broader context, such an approach will guide our efforts to study the effects of genetic metabolic disorders, malnutrition, and maternal diabetes on embryonic development.

One of the earliest cell fate decisions is the exit from pluripotency, resulting in the emergence of the three germ layers: the ectoderm, mesoderm, and endoderm. In the embryo, prospective mesodermal and endodermal cells migrate through a transient structure known as the primitive streak (PS), while remaining epiblast cells adopt an ectodermal cell fate.[19,20] Much attention has focused on how a cell's preference for either aerobic glycolysis or oxidative phosphorylation (OxPhos) changes during differentiation.[18] Pluripotent stem cells (PSCs) are thought to rely on high glycolytic activity to maintain their characteristic histone acetylation patterns.[21] Although some studies have suggested that metabolic switching is a prerequisite for epigenetic remodeling and differentiation,[21,22] others have found that a shift toward oxidative metabolism is germ-layer specific and only occurs in mesodermal and endodermal cells.[23] In contrast to these findings from directed differentiation of cells

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using extrinsic signals, a study in the developing tailbud (neuromesodermal progenitors [NMPs]) found that inhibition of glycolysis increased the proportion of neuroectoderm at the expense of the presomitic mesoderm (PSM) via regulation of Wnt signaling.[15] Unlike the bipotent NMPs, cells of the early embryo can still give rise to all future cell types. Therefore, the relationship between metabolism and signaling during the earliest stages of differentiation of pluripotent embryonic cells into the three germ layers remains unresolved. Recent studies, both *in vitro* and *in vivo*, have reported spatiotemporal restriction of different glucose (Glc) transporters that accompany germ layer patterning and how metabolism affects signaling, particularly during mesoderm specification.[16,24] However, the simultaneous control of the three germ layers and their cell-type-specific interplay between glycolysis and signaling remains obscure.

In this study, we further elucidate the interplay between glycolysis and the signaling pathways that coordinate germ layer differentiation. Using 3D mouse gastruloids, a stem cell-based model system that allows the co-differentiation of the three germ layers,[25] we found that glycolysis plays a crucial role in both endoderm and mesoderm induction by acting as a positive regulator of Nodal, Wnt, and Fgf signaling. Importantly, exogenous Glc has a dose-dependent effect on PS marker expression and the development of endodermal and mesodermal cell type derivatives. Thus, we show that glycolytic activity acts as an upstream regulator that can instruct germ layer proportions. By developing a novel genetic tool to manipulate Glc availability in mammalian cells, we also demonstrate that increased glycolytic activity induces Brachyury expression and promotes gastruloid development in a time-dependent manner. Moreover, we were able to rescue mesoderm and endoderm development in glycolysis-inhibited gastruloids with agonists of the Nodal and Wnt signaling pathways. This revealed that the effects of glycolytic activity on germ layer specification can be decoupled from its effects on gastruloid growth. Therefore, it seems that glycolytic activity is not a bioenergetic prerequisite for endoderm and mesoderm differentiation but primarily functions as an important positive regulator of Nodal and Wnt signaling. These findings demonstrate how glycolytic activity acts as a regulator of morphogen signaling and cell fate determination and open new possibilities for metabolic control of cell type proportions in *in vitro* systems.

RESULTS

Glycolysis is needed for T/Bra expression and symmetry breaking in gastruloids

To address the role of metabolism during germ layer specification, we used gastruloids, aggregates of mouse embryonic stem cells (mESCs) that specify cell types of all three germ layers while establishing an anterior-posterior (AP) axis.[25] Between 24 and 48 h post aggregation (hpa), gastruloids upregulate the PS and early mesoderm marker *Brachyury* (*T/Bra*)[26,27] throughout the tissue (Figure 1A), therefore approximately linking to the mouse embryo at E6.0/E6.5. In a symmetry-breaking event, the posterior pole is then specified by the polarization of *T/Bra* expression. At 72 hpa, markers of all three germ layers can be detected and further culture results in gastruloid elongation along the AP axis.[28] To provide a clean delineation between signaling and metabolic activity, and in contrast to previous culture protocols, we did not add CHIR99021 (CHIR, glycogen synthase kinase-3 [GSK3] inhibitor and Wnt signaling activator) to the differentiation medium. Even in the absence of CHIR treatment, we found that ~95% of gastruloids generated from E14 T/Bra::GFP mESCs break symmetry spontaneously and elongate when they were previously maintained in serum/LIF conditions (Figure 1A).

To investigate the role of Glc metabolism during germ layer specification, we used 2-deoxy-D-glucose (2-DG)[29] to inhibit glycolysis in gastruloids (Figure S1A). Using the T/Bra::GFP reporter line, we found that 2-DG treatment between 24 and 48 hpa resulted in the loss of T/Bra::GFP expression and subsequent failure of symmetry breaking and axis elongation (Figure 1A). Time-lapse imaging showed upregulation of T/Bra::GFP expression in control gastruloids between 24 and 48 hpa, whereas 2-DG-treated gastruloids did not start expressing the PS marker, even after 2-DG withdrawal (Figure 1B; Videos S1 and S2). The effect on T/Bra::GFP expression was dependent on 2-DG concentration (Figure S1B). In addition, we found that other inhibitors of aerobic glycolysis (3-bromopyruvate [3-BP] and dichloroacetate [DCA]) had similar dose-dependent effects on gastruloid development (Figures S1A–S1C). To exclude any side effects of the inhibitors, we next aimed to remove Glc directly from the medium. We prepared N2B27 differentiation medium with 0.5 mM Glc, reflecting physiological Glc levels in the uterine fluid.[30] Although this is around 40 times less than the Glc concentration used in commercially available N2B27 medium, T/Bra::GFP expression and symmetry breaking during gastruloid development were robust (Figure 1C). Additionally, the relatively low concentration allowed us to remove Glc (<0.004 mM Glc) between 24 and 48 hpa of gastruloid development by repeated washes with Glc-free medium. Consistent with our hypothesis that the observed 2-DG phenotype was mediated by reduced glycolytic activity, lack of Glc in this time window also resulted in reduced T/Bra::GFP upregulation and failure to polarize (Figure 1C). This phenotype could be rescued by the addition of galactose, fructose, and mannose between 24 and 48 hpa (Figures S2A–S2E). These monosaccharides can enter the glycolytic pathway at different steps and were all able to restore lactate secretion and, therefore, glycolytic activity (Figures S2A and S2D). In contrast, the downstream metabolite pyruvate was neither able to rescue glycolytic activity nor gastruloid development (Figures S2D and S2E). These experiments suggest that glycolytic pathway activity is needed during early gastruloid development to ensure T/Bra expression and subsequent symmetry breaking.

To ask whether T/Bra::GFP expression is also influenced by other metabolic pathways, we next treated gastruloids with sodium azide (NaN_3), a complex IV inhibitor that blocks OxPhos (Figures S3A and S3B). Although the treatment resulted in an effective block of OxPhos, as measured by loss of oxygen consumption as well as reduced gastruloid size, we did not observe loss of T/Bra::GFP expression and symmetry breaking (Figures S3C and S3D). We further validated this finding by inhibiting OxPhos with complex I (rotenone) and complex III (antimycin A) inhibitors (Figure S3D). These results suggest that metabolic inhibition and reduced tissue growth do not affect T/Bra::GFP expression per se and indicate that glycolysis might play a specific role in gastruloid development.

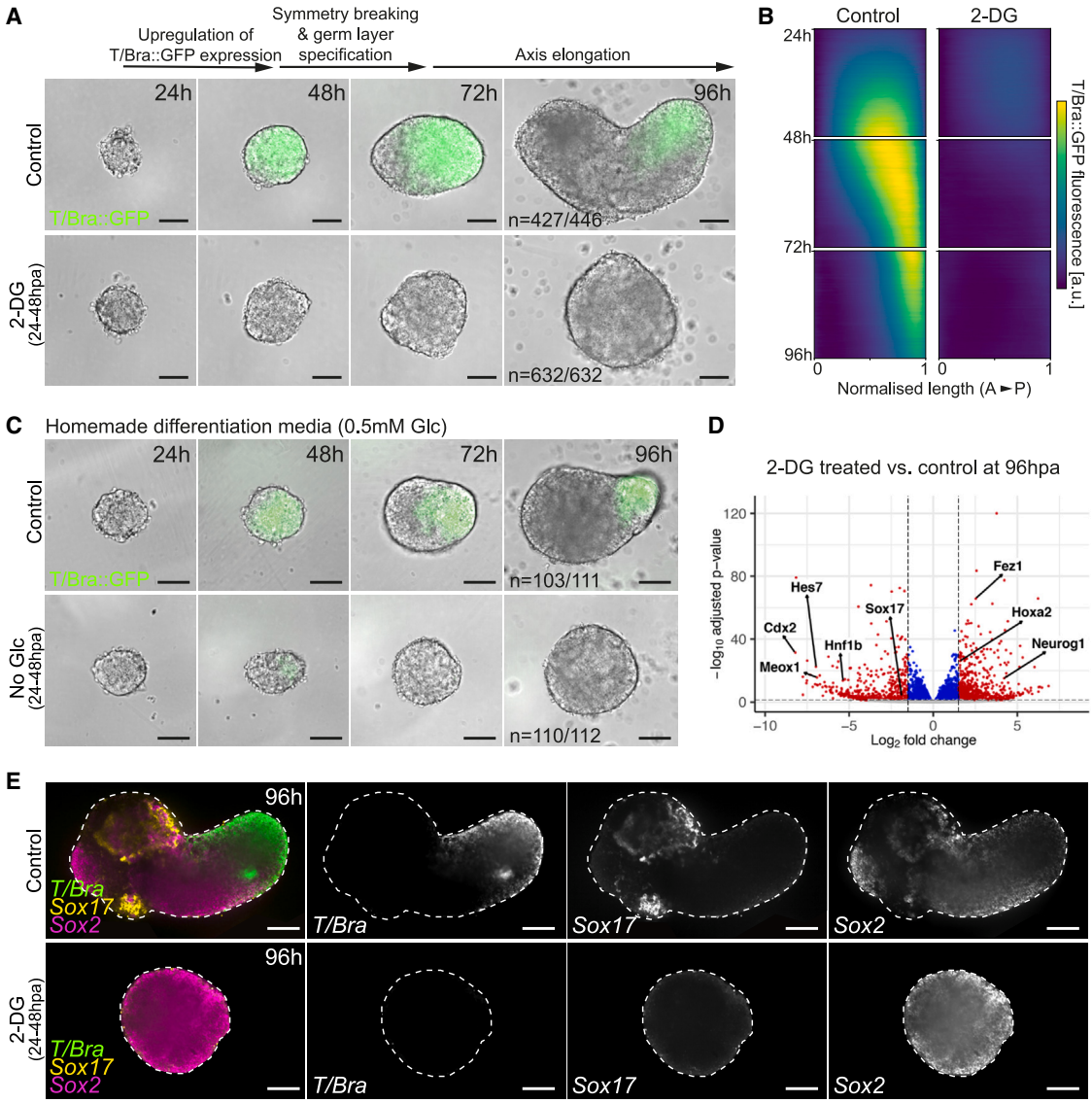


Figure 1. Glycolysis is required for T/Bra expression and mesoderm and endoderm induction

(A) Glycolysis inhibition between 24 and 48 hpa results in loss of T/Bra::GFP expression and impaired gastruloid development. *N*_exp = 12. Scale bars: 100 μm.

(B) Kymographs of control and 2-DG-treated gastruloids. 2-DG-treated gastruloids fail to upregulate and polarize T/Bra::GFP expression. Normalized length from anterior (A) to posterior (P). *n* = 30 for each condition.

(C) Gastruloids in differentiation media containing 0.5 mM Glc express T/Bra::GFP and break symmetry. Removing Glc between 24 and 48 hpa results in loss of T/Bra::GFP expression. *N*_exp = 3. Scale bars: 100 μm.

(D) Volcano plot describing differentially expressed genes between 2-DG-treated and control gastruloids at 96 hpa based on RNA-seq. Red dots represent genes that pass thresholds for $\log_2$FC = ±1.5 (vertical lines) and adjusted *p* value = 0.05 (horizontal line). *n* = 6 for both conditions.

(E) HCR stainings of control and 2-DG-treated gastruloids at 96 hpa. Single confocal slices are shown. *N*_exp = 3, *n* = 107 (control), *n* = 104 (2-DG). Scale bars: 100 μm.

See also Videos S1 and S2.

2-DG suppresses mesoderm and endoderm formation

Gastruloids display post-occipital cell types of all three germ layers, which are spatially organized along an AP axis.[31,32] We next asked how the early inhibition of glycolysis affects cell fate decision-making during gastruloid development. To assess the presence of different cell types, we performed single-gastruloid bulk RNA sequencing (RNA-seq) on control and 2-DG-treated gastruloids at 96 hpa. We found that markers of mesodermal derivatives, such as *Hes7* and *Meox1*, were strongly downregulated[33,34] (Figures 1D and S4A). Similarly, endodermal markers, such as *Sox17 and Hnf1b*, were reduced in 2-DG-treated gastruloids[35] (Figures 1D and S4B). In contrast, transcripts associated with neural cell fates, such as *Neurog1* and *Fez1*, were upregulated[36,37] (Figures 1D and S4C). Furthermore, we found an upregulation of the most anterior *Hox* gene and hindbrain marker *Hoxa2*, whereas posterior Hox genes

and *Cdx2*, which marks the posterior embryo, were strongly reduced[38,39] (Figures 1D and S4E). This suggests an anteriorization of 2-DG-treated gastruloids.

The shift in germ layer proportions was further confirmed by *in situ* hybridization chain reaction (HCR), showing that, in contrast to control gastruloids, cells expressing *T/Bra* (PS/early mesoderm) and *Sox17* (definitive endoderm) transcripts were strongly reduced in 2-DG-treated gastruloids (Figure 1E). Instead, *Sox2* (ectoderm/pluripotent) was expressed throughout glycolysis-inhibited gastruloids. Further stainings showed that the *Sox2*$^+$ domain was subdivided into clusters of cells expressing the neuroectodermal marker *Sox1* or the pluripotency marker *Nanog*[40,41] (Figure S4F). Both of these markers were overrepresented compared with control gastruloids, confirming the transcriptomics results (Figures S4C and S4D).

These results show that the inhibition of glycolysis between 24 and 48 hpa has long-term effects on gastruloid development and results in a shift of germ layer proportions away from posterior mesodermal and endodermal derivatives toward more anterior neuroectodermal cell fates.

Glc has a dose-dependent effect on the proportion of T/Bra::GFP-expressing cells

The observed effects of 2-DG treatment on gastruloid development raise the question of how the level of glycolytic activity relates to *T/Bra* expression and germ layer induction. One possibility is that a certain threshold of cellular glycolytic activity is needed to enable *T/Bra* upregulation in gastruloids. The other possibility is that glycolytic activity has a dose-dependent effect on *T/Bra* expression, as seen for instructive signals such as morphogens.[42] To address this question, we aimed to modulate the glycolytic activity in gastruloids to assess the effect on T/Bra::GFP expression.

As a first step, we determined the concentration range in which exogenous Glc affects the glycolytic activity of cells cultured in 2D. By performing real-time measurements of extracellular acidification and oxygen consumption, we found that the glycoATP production rate (ATP production rate from glycolysis) rapidly increased with rising exogenous Glc levels before plateauing around 12.5 mM Glc (Figures 2A and S5A). In the absence of Glc, cells relied entirely on OxPhos for ATP production (Figure S5B). Raising Glc levels resulted in increased glycolytic activity relative to OxPhos, suggesting a gradual change of the metabolic profile. At 12.5 mM Glc, glycolysis and OxPhos contributed approximately evenly to the total ATP production rate (Figure S5B). We further confirmed these changes in glycolytic activity by measuring secreted lactate levels (product of aerobic glycolysis) and the NAD$^+$/NADH ratio (decreases with increased glycolytic activity) in cells cultured at different Glc concentrations (Figures S5C and S5D).

We next generated gastruloids in medium containing 0.02, 0.1, 0.5, and 2.5 mM Glc, the range where changes in glycolytic activity were most pronounced (Figure S5A). At 48 hpa, when T/Bra::GFP expression is present throughout the gastruloid, a positive correlation between Glc concentration and GFP signal was observed (Figure 2B). However, it was also apparent that gastruloids grown in lower Glc concentrations were reduced in size, suggesting that Glc was growth limiting. To exclude the possibility that size and not glycolytic activity is affecting T/Bra::GFP expression, we generated gastruloids from different initial cell numbers, resulting in similar sized gastruloids at different Glc concentrations (Figure 2C, see inlets). By assessing the percentage of GFP$^+$ cells using flow cytometry over four different initial cell numbers (350, 700, 1,050, and 1,400 cells), we found that the percentage of GFP$^+$ cells was indeed dependent on Glc concentration and not gastruloid size (Figures 2C and S5E). We further wondered whether Glc concentration not only determines the percentage of GFP$^+$ cells but also cellular T/Bra::GFP expression levels. Based on flow cytometry histograms, we detected no noticeable shift in typical GFP peak intensity levels, suggesting that cellular T/Bra::GFP expression levels are similar in GFP$^+$ cells across different Glc concentrations (Figure S5F). Exogenous Glc concentration thus determines the likelihood of a cell to switch into a *T/Bra*-expressing state in a dose-dependent manner. As a result, gastruloids developing in higher Glc concentration display a higher percentage of T/Bra::GFP$^+$ cells at 48 hours post fertilization (hpf), resulting in a higher overall T/Bra::GFP intensity level. These results indicate that glycolytic activity plays a regulative role in *T/Bra* induction during gastruloid development, especially between 0 and 0.5 mM Glc where dose-dependency is most prominent (Figure S5E).[29]

Glc has a dose-dependent effect on germ layer proportions

Next, we asked whether the dose-dependent effect of Glc on the relative number of T/Bra::GFP-expressing cells further translates into a change in the proportion of cell types derived from the different germ layers. We first determined the dynamics of T/Bra::GFP reporter activity at different Glc concentrations by performing time-lapse imaging. Kymographs of T/Bra::GFP reporter intensity showed that gastruloids developing at 0.02 mM Glc failed to polarize (Figure 2D). Instead, a weak stripe of T/Bra::GFP expression formed—a phenotype previously described in gastruloids treated with Nodal or Fgf signaling inhibitors.[43] Despite differences in reporter intensities, the dynamics of symmetry breaking at 0.1, 0.5, and 2.5mM Glc were comparable (Figure 2D).

Next, we performed HCR stainings for *Sox2* and markers of PS-derived cell fates at 72 hpa. We found that the proportion of *Sox2*-expressing cells decreased with rising Glc concentration (Figures 2E and 2F). In contrast, mesodermal and endodermal markers, such as *Eomes* (endoderm/anterior mesoderm), *Gata6* (cardiac mesoderm/endoderm), *Aldh1a2* (trunk mesoderm), and *Meox1* (paraxial mesoderm), were absent in gastruloids cultured at 0.02 mM Glc but showed an increase in proportion with rising Glc concentration (Figures 2E and 2F). Our results show that exogenous Glc levels can bias the emergence of specific cell types in a dose-dependent manner. Thus, increasing Glc levels lead to a greater proportion of mesodermal and endodermal cell derivatives, accompanied by lower proportions of ectodermal cell fates. This demonstrates how Glc concentration can be used as a tool to modulate the proportions of cell lineages in gastruloids.

A Glc-producing cell aggregate can rescue gastruloid development in a time-dependent manner

The targeted manipulation of individual metabolite levels and metabolic pathway activity is crucial for testing their importance

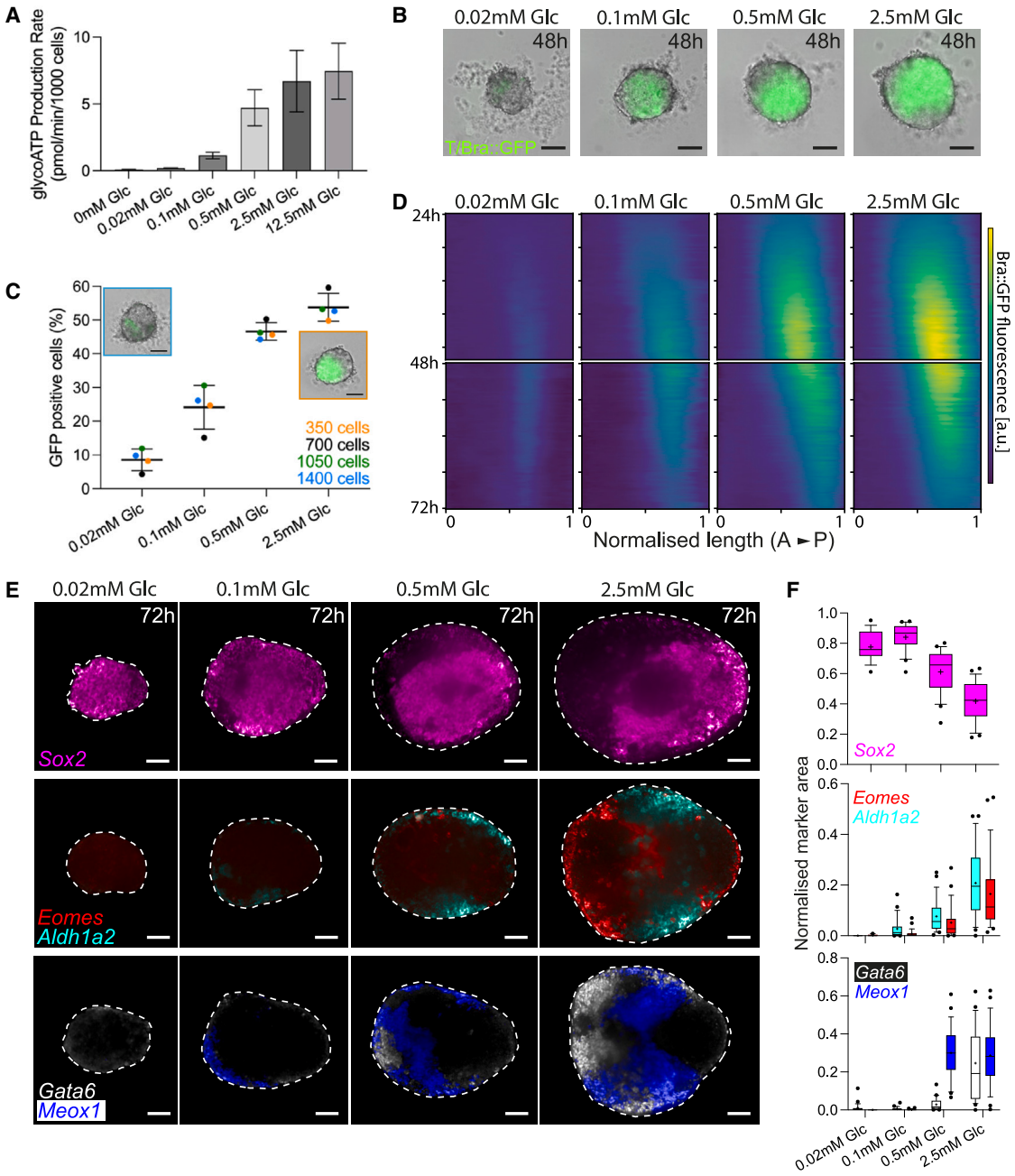


Figure 2. Glc has a dose-dependent effect on T/Bra::GFP-expressing cells and germ layer proportions

(A) Mean glycoATP production rate depending on exogenous Glc levels. Error bars indicate SD. *N*_exp = 3 for all conditions.

(B) Gastruloids grown in increasing exogenous Glc concentrations show increasing levels of T/Bra expression. *N*_exp = 7, $n \geq 315$ for each condition. Scale bars: 100 μm.

(C) Mean percentage of GFP$^+$ cells in T/Bra::GFP gastruloids at 48 hpa, which were generated by aggregating 350, 700, 1,050, and 1,400 cells and cultured in different Glc concentrations. Error bars indicate SD. Inlets show images of gastruloids generated by aggregation of 1,400 (blue frame) and 350 (orange frame) initial cells and grown in 0.02 and 2.5 mM Glc, respectively. At 48 hpa, gastruloids are similar sized but express different levels of T/Bra::GFP. Scale bars: 100 μm.

(D) Kymographs of T/Bra::GFP intensity in gastruloids developing in different Glc concentrations from 24 to 72 hpa. *n* = 10 for each condition.

(E) Germ layer marker expression in gastruloids at 72 hpa, which developed at different Glc concentrations. Single confocal slices are shown. *N*_exp = 3, $n > 31$ for each condition. Scale bars: 50 μm.

(F) Germ layer marker area normalized to gastruloid area for the batch of gastruloids shown in (E). Line in boxplot indicates median, + indicates mean, whiskers indicate 10th to90th percentiles. *Sox2*: *n* = 18 (0.02 mM Glc), *n* = 22 (0.1 mM Glc), *n* = 23 (0.5 mM Glc), *n* = 22 (2.5 mM Glc). *Eomes/Aldh1a2*: *n* = 18 (0.02 mM Glc), *n* = 26 (0.1 mM Glc), *n* = 25 (0.5 mM Glc), *n* = 25 (2.5 mM Glc). *Gata6/Meox1*: *n* = 10 (0.02 mM Glc), *n* = 18 (0.1 mM Glc), *n* = 22 (0.5 mM Glc), *n* = 20 (2.5 mM Glc).

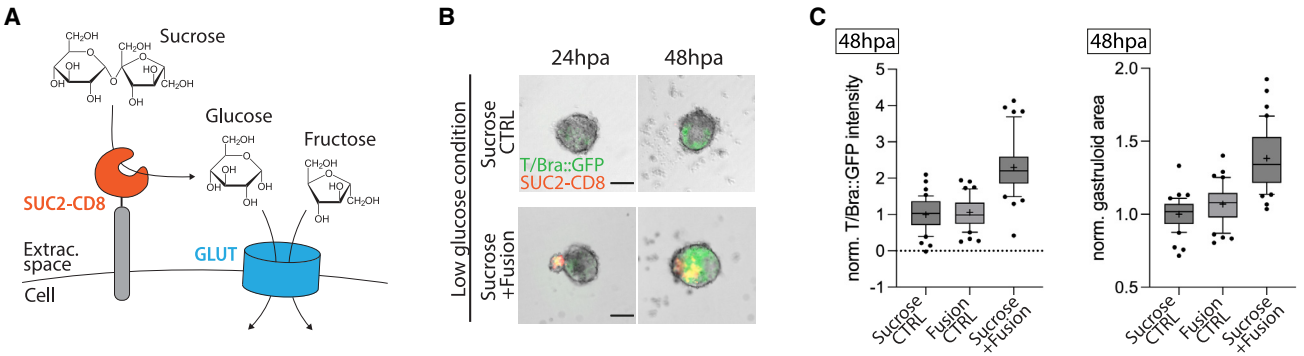


Figure 3. A Glc-producing cell aggregate can rescue gastruloid development in a time-dependent manner

(A) Schematic explaining how the expression of SUC2-CD8 allows cells to metabolize sucrose.

(B) Upper row (sucrose CTRL): gastruloid development in low-Glc conditions, 10 mM sucrose added from 26 hpa onward. Lower row (sucrose + fusion): fusion of SUC2-CD8-expressing aggregate rescues T/Bra::GFP expression and growth of the entire gastruloid in the presence of sucrose. Scale bars: 100 μm. Complete time course in Figure S6E.

(C) Quantifications of T/Bra::GFP expression (left) and gastruloid area (right) at 48 hpa. Values are normalized to sucrose CTRL. Sucrose CTRL: no fusion but 10 mM sucrose from 26 hpa onward. Fusion CTRL: fusion at 24 hpa but no added sucrose. Sucrose + fusion: fusion at 24 hpa, 10 mM sucrose from 26 hpa onward. Line in boxplot indicates median, + indicates mean, whiskers indicate 10th to 90th percentiles. *N*_exp = 3, *n* = 42 (sucrose CTRL), *n* = 41 (fusion CTRL), *n* = 42 (sucrose + fusion).

during cellular and developmental processes. However, designing genetic tools to manipulating metabolism can be challenging due to the complex regulation of metabolic fluxes. Currently, there is still a shortage of such tools in mammalian cells.[44] Thus, we aimed to develop a novel tool to increase glycolytic activity in cells to further validate its importance during gastruloid development.

Due to the lack of specific transporters and enzymes, mammalian cells are not able to utilize sucrose, a disaccharide composed of Glc and fructose subunits. SUC2 is a yeast invertase that can cleave sucrose into Glc and fructose.[45,46] By membrane-tethering SUC2 (SUC2-CD8) on the outside of the cell surface, we engineered mESCs to be glycolytically active in Glc-free medium with sucrose supplementation (Figures 3A, S6A, and S6B). Although gastruloids upregulate T/Bra::GFP expression and undergo symmetry breaking in high-Glc conditions, gastruloid development is impaired in low-Glc conditions (Figure 2D). Thus, we assessed whether the fusion of a small aggregate expressing SUC2-CD8 to a gastruloid would rescue its development in low-Glc condition (0.02 mM Glc) upon sucrose addition (Figure S6C). Indeed, fusing SUC2-CD8-expressing cells at 24 hpa rescued T/Bra::GFP expression as well as gastruloid growth upon sucrose addition (Figures 3B, 3C, and S6D). Furthermore, we found that the timing of glycolytic activity was crucial for gastruloid development because fusion at 48 hpa was not sufficient to rescue T/Bra::GFP expression to the same extent (Figure S6E). Together, these experiments show that a Glc-producing aggregate can induce T/Bra::GFP expression and rescue gastruloid development in a time-dependent manner.

Nodal, Wnt, and Fgf signaling pathway activity is dependent on glycolysis

We next sought to understand the mechanism by which glycolytic activity regulates changes in cell fate decision-making. PS markers are induced and maintained by major developmental signaling pathways, such as Nodal, Wnt, and Fgf signaling.[19,47–49] To test whether the effect of glycolysis inhibition on mesoderm and endoderm development is mediated by reduced signaling, we performed single-gastruloid RNA-seq on control and 2-DG-treated gastruloids immediately at the end of the treatment period at 48 h of gastruloid development. We found that transcription of Nodal, Wnt3a, and Fgf8 ligands, as well as many of their target genes such as *T/Bra*, *Eomes*, and *Dusp6*, were downregulated after glycolysis inhibition (Figure 4A). Among the most upregulated transcripts were *Meis2*, *Irx3*, and *Chrdl1*, all genes associated with neural development[50–52] (Figure S7A). These results suggest that the effect of reduced glycolytic activity on germ layer proportions can be explained by the reduced activity of Nodal, Wnt, and Fgf signaling.

Development of glycolysis-inhibited gastruloids can be rescued by Nodal or Wnt signaling pathway activation

The observation that Nodal, Wnt, and Fgf signaling pathways are downregulated upon glycolysis inhibition raises the question of whether mesoderm and endoderm development are impaired because of the absence of morphogen signaling. If glycolysis is primarily required for the activation of these pathways or to promote cells required to induce these signaling pathways, it should be possible to rescue the 2-DG phenotype by activating signaling pathways during the treatment period. Another possibility is that mesoderm and endoderm specification cannot take place under the bioenergetic constraints related to the lack of glycolytic activity. In this case even the activation of the signaling pathways should not be able to rescue the phenotype. We found that the simultaneous activation of the Nodal signaling pathway with activin A (Act-A) during 2-DG treatment between 24 and 48 hpa was able to partially rescue T/Bra::GFP expression at 48 hpa (Figures 4B and 4C). Even though T/Bra::GFP expression did not reach control levels at 48 hpa, symmetry breaking was robust and elongation was observed in ~70% of the cases. We further performed HCR stainings at 96 hpa to validate whether cells of all three germ

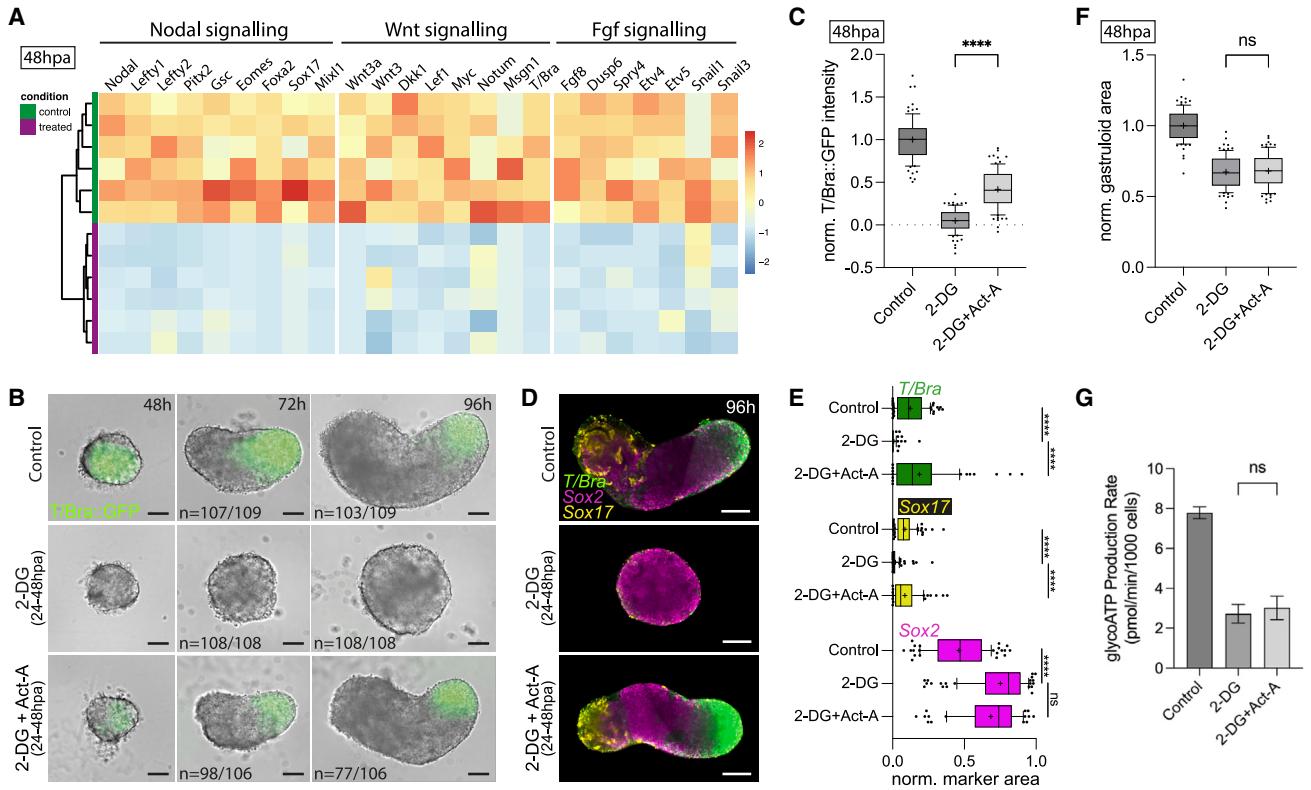


Figure 4. Nodal signaling activation can rescue the 2-DG phenotype in the absence of glycolytic activity

(A) Heatmap depicting relative transcript levels of selected genes in 2-DG-treated vs. control gastruloids at 48 hpa. Heatmap columns are normalized per gene. Lower and higher expression in blue and red, respectively. Each row represents a single gastruloid.

(B) Gastruloid development under different conditions. n numbers for symmetry breaking at 48 hpa and axis elongation at 96 hpa from *N*_exp = 4. Scale bars: 100 μm.

(C) T/Bra::GFP intensity of gastruloids at 48 hpa for different treatment conditions normalized to mean control intensity. Line in boxplot indicates median, + indicates mean, whiskers indicate 10th to 90th percentiles. *N*_exp = 3, *n* = 87 (control), *n* = 85 (2-DG), *n* = 87 (2-DG + Act-A). ****$p < 0.0001$ for t test.

(D) Images of HCR stainings of gastruloids at 96 hpa under different conditions. Single confocal slices are shown. Scale bars: 100 μm.

(E) Germ layer marker area normalized to gastruloid area of images, as shown in (D). Line in boxplot indicates median, + indicates mean, whiskers indicate 10th to 90th percentiles. *N*_exp = 3, *n* = 107 (control), *n* = 104 (2-DG), *n* = 70 (2-DG + Act-A). ****$p < 0.0001$ for t test.

(F) Normalized gastruloid area at 48 hpa. Line in boxplot indicates median, + indicates mean, whiskers indicate 10th to 90th percentiles. *N*_exp = 3, *n* = 87 (control), *n* = 85 (2-DG), *n* = 87 (2-DG + Act-A). t test was not significant (ns).

(G) Mean glycoATP production rate of cells cultured for 48 h in differentiation medium. Inhibitor and growth factor treatment was started 24 h prior to measurements. *N*_exp = 3, error bars indicate SD. t test was not significant (ns).

layers can be found in the rescued gastruloids. Quantifications of the relative area of *T/Bra* and *Sox17* expression suggested that mesoderm and endoderm development were rescued (Figures 4D and 4E). Further stainings also confirmed the presence of paraxial mesoderm (*Meox1*) and endoderm/anterior mesoderm (*Gata6*) in these gastruloids (Figure S7B). These results confirm that Nodal signaling activation during glycolysis inhibition was sufficient to rescue the development of mesodermal and endodermal cell types. Activation of Wnt signaling using CHIR was also able to rescue gastruloid development, albeit in a less-reproducible manner (Figures S7C and S7D). Even though elongation occurred in ∼34% of gastruloids treated with CHIR, endodermal and cardiac cell derivatives were mostly absent (Figures S7B and S7E). We also added Fgf8 simultaneously with 2-DG but did not observe a rescue of T/Bra::GFP induction and polarization (Figure S7F). Our rescue experiments suggest that Nodal and Wnt signaling activation are downstream of glycolytic activity and can rescue the 2-DG-mediated phenotype on gastruloid development.

Glycolytic activity acts as an activator of Nodal and Wnt signaling in micropatterned human gastruloids

To further investigate whether these findings are relevant for human development, we used micropatterned human ESC (hESC) colonies, which develop radial patterns of all three germ layers upon BMP4 treatment.[53] In human gastruloids, PS-like cells expressing T/BRA emerge around 24 h post BMP4 induction (hpi), therefore approximately linking to the human embryo at Carnegie collection stage 6b.[53–56] Using both ESI017 and H9 hESCs, we found that 2-DG treatment between 24 and 48 h post differentiation induction with BMP4 (hpdi) resulted in loss of T/BRA and increased levels of SOX2 at 48 hpdi (Figures 5 and S7G–S7M). In accordance with the observation that Nodal signaling is downregulated upon glycolysis inhibition in mESCs,

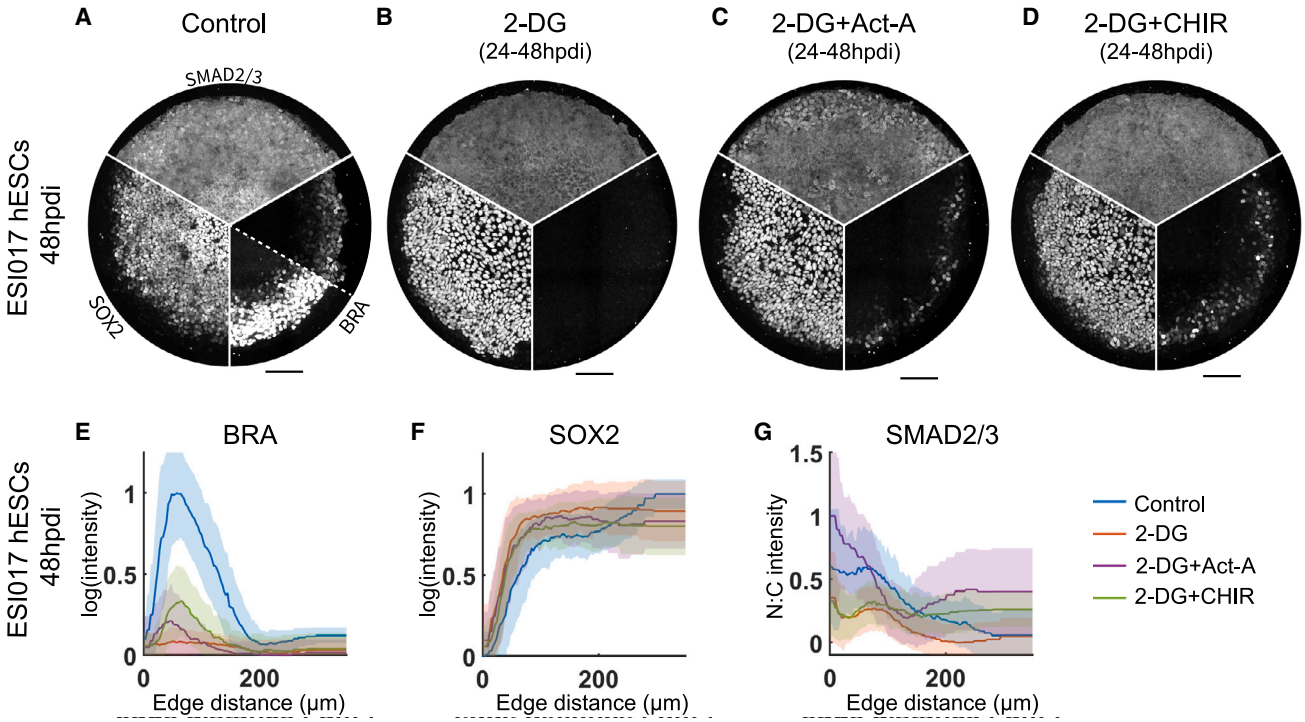


Figure 5. Nodal and Wnt signaling activation can partially rescue T/BRA expression in glycolysis-inhibited human gastruloids

(A–D) Immunofluorescence images of micropatterned ESI07 human gastruloids at 48 hpdi, with control BMP treatment from 0 to 48 hpdi (A), combined with 2-DG treatment from 24 to 48 hpdi (B), 2-DG + Act-A from 24 to 48 hpdi (C), and 2-DG + CHIR from 24 to 48 hpdi (D). Panels have the same contrast limits, but because T/BRA intensity in control (A) is very high compared with other conditions, we show a non-saturated image in the pie above the white dotted line. Scale bars: 50 μm. All colonies are 700 μm in diameter.

(E–G) Spatial distribution of T/BRA (E) and SOX2 (F) log(intensity), and nuclear-cytoplasmic intensity ratio of Smad2/3 (Act-A/Nodal signaling) (G), averaged over $n = 3$ colonies. Dark lines represent the mean over colonies, colored bands represent the standard deviation.

nuclear/cytoplasmic levels of SMAD2/3 were reduced upon 2-DG treatment in micropatterned hESCs colonies (Figures 5G and S7M). Furthermore, the addition of Act-A or CHIR was able to partially rescue T/BRA expression (Figures 5C–5G and S7I–S7M). These results suggest that the role of glycolytic activity as a positive regulator of the PS marker T/BRA via Nodal and Wnt signaling is conserved in human cells.

Glycolytic activity is not rescued by Nodal and Wnt signaling

The signaling, epigenetic, and metabolic states of cells are highly integrated through several feedback mechanisms that ensure coordination of cellular behavior.[1,5,7,57] In cancer cells, Wnt signaling is a crucial regulator of metabolic reprogramming, resulting in increased Glc uptake and its preferential fermentation to lactate even in the presence of oxygen, known as the Warburg effect.[58] Similarly, Wnt, Fgf, and Nodal signaling have been suggested to promote glycolysis in the context of normal embryonic development and homeostasis as well as in cancerous tissue.[14,59–62]

To investigate the possibility that the rescue of 2-DG-treated gastruloids by Act-A and CHIR was the result of signaling-mediated activation of glycolysis, we first analyzed gastruloid size, which is expected to be affected by 2-DG treatment.[63] Indeed, we detected a significant reduction in gastruloid size after the 24-h 2-DG treatment period. In contrast to T/Bra::GFP intensity levels, neither Nodal nor Wnt signaling activation rescued the gastruloid size phenotype, suggesting that anabolic metabolism driving proliferation and tissue growth was not restored in these gastruloids (Figure 4F; Figure S7N). To look at a more direct readout of metabolic state, we next measured the ATP production rate from glycolysis (glycoATP production rate). Control cells cultured for 2 days in differentiation medium displayed a significantly higher glycoATP production rate than cells that had been treated with 2-DG for 24 h prior to measurement (Figure 4G). We found that the addition of Nodal signaling agonist Act-A did not rescue the glycoATP production rate of 2-DG-treated cells. This was also the case when Act-A and CHIR were added simultaneously to cells (Figure S7O). These results were further supported by the finding that T/Bra::GFP expression and symmetry breaking of gastruloids in Glc-free medium (24–48 hpa) was also rescued by the addition of Act-A and CHIR. (Figure S7P). Here, the absence of Glc abolishes any possibility of reactivating glycolytic activity. Together, these experiments suggest that even though signaling activation can rescue gastruloid development, it does not rescue glycolysis in 2-DG-treated cells. This implies that glycolytic activity is not needed for endoderm and mesoderm specification as long as signaling pathways are active. Therefore, the influence of glycolysis on mesodermal and endodermal cell fates can be decoupled from its requirement to support the bioenergetic needs of cells.

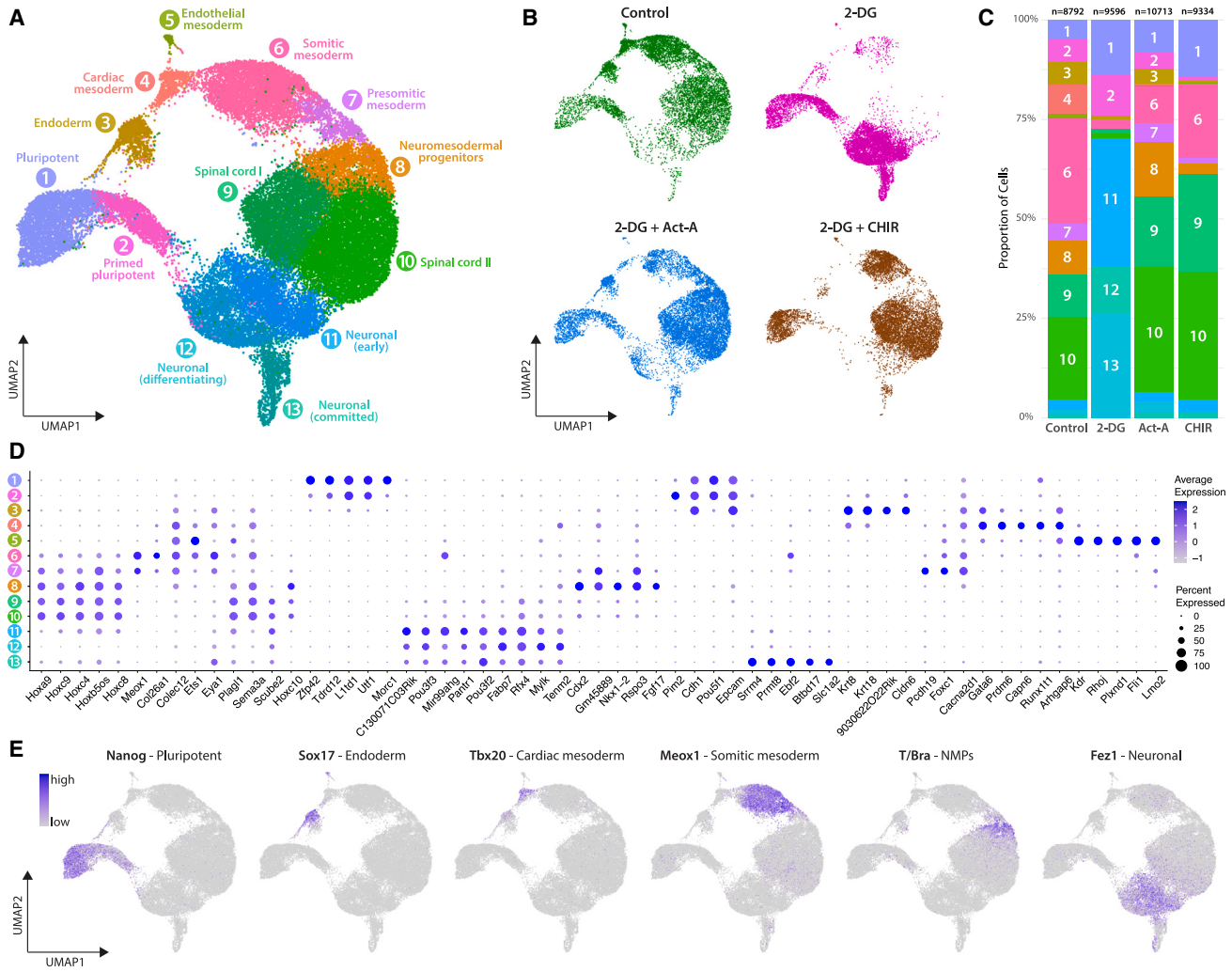


Figure 6. scRNA-seq reveals cell-type-specific rescue by Nodal and Wnt signaling activation

(A) UMAP plot showing the combined cell populations identified in control, glycolysis-inhibited (2-DG), and rescued (2-DG + Act-A, 2-DG + CHIR) gastruloids at 96 hpa. Cells are colored and numbered by their cluster annotation.

(B) UMAP plots visualizing the abundance of different cell types in gastruloids of each condition.

(C) Stacked bar plot representation of cell populations from (A) shown as percentage of total analyzed cell numbers and separated by condition. Colors correspond to clusters specified in (A). Cell types contributing >3% are numbered according to their cluster. All percentages and cell numbers are listed in Figure S9B. Total analyzed cell number: n = 8,792 (control), n = 9,596 (2-DG), n = 10,713 (2-DG + Act-A), n = 9,334 (2-DG + CHIR).

(D) Dot plot of the Top5 differentially expressed genes of each cluster (see Table S1).

(E) Feature plots with the expression of key marker genes in the integrated UMAP.

See also Table S1.

scRNA-seq confirms rescue of early and late PS populations by Nodal signaling in glycolysis-inhibited gastruloids

To determine whether all endodermal and mesodermal cell fates can be rescued by signaling activation in glycolysis-inhibited gastruloids, we next performed single-cell RNA sequencing (scRNA-seq) analysis on control, 2-DG-treated, and rescued (by the addition of Act-A and CHIR) gastruloids at 96 hpa. In accordance with our previous bulk RNA-seq data (Figure 1D), the scRNA-seq analysis confirmed the drastic reduction of anterior mesoderm (cardiac) and posterior mesodermal (presomotic and somitic) populations, as well as endodermal cell populations in glycolysis-inhibited gastruloids (Figure 6, and S8; Table S1). In addition, we found striking differences in the neuroectodermal cell populations of control and 2-DG-treated gastruloids. Control gastruloids contained predominantly cells with spinal-cord-like identity derived from NMPs. In contrast, glycolysis-inhibited gastruloids were depleted of NMPs and spinal-cord-like cells and instead consisted largely of more anterior neuroectoderm, characterized by markers for different stages of neuronal differentiation and the lack of posterior Hox genes (Figures 6B, 6C, S9A, and S9B). Mapping of these neuronal clusters onto a mouse embryo scRNA-seq dataset[64] suggested a closer relation to the forebrain/midbrain/hindbrain cluster than to any other neuroectodermal cluster

(Figure S9C). In addition, the proportion of pluripotent and primed pluripotent cells was increased (control: 10.5%, 2-DG: 24.1%), suggesting that glycolysis inhibition also affected exit from pluripotency (Figure S8B).

The addition of the Nodal signaling agonist Act-A was able to rescue all cell types present in the control, with the exception of the cardiac mesoderm (Figures 6B, 6C, and S8B). One explanation could be that there is an energetic requirement of glycolytic activity during the specification of cardiac progenitors. Likewise, it could be that the Act-A treatment shifts cell fates toward endoderm differentiation because cardiac differentiation requires more moderate levels of Nodal signaling.[65]

The Wnt signaling agonist CHIR neither rescued endoderm nor cardiac mesoderm. This is most likely related to the importance of Nodal signaling for definitive endoderm and cardiac mesoderm, which are among the earliest cell types that ingress through the PS.[19] It is conceivable that the strong activation of Wnt signaling drives cells toward more posterior cell fates. In addition, the overrepresentation of somitic-mesoderm-like and spinal-cord-like cells in the absence of NMPs and PSM cells suggests a depletion of these progenitor cell types at earlier stages of gastruloid development (Figures 6B and 6C).

The scRNA-seq analysis confirmed that the inhibition of glycolysis with 2-DG strongly reduces all cell types that are derived from the PS, while more anterior neuronal populations are promoted. The addition of Nodal and Wnt signaling agonists during glycolysis inhibition does not only rescue gastruloid development on the morphological level but can also restore cell types derived from the PS. In particular, Nodal signaling activation restores both early and late PS populations.

DISCUSSION

A crucial role of glycolytic activity for mesoderm and endoderm induction

Spatial metabolic patterns observed during amphibian gastrulation in the 1930s sparked early hypotheses about the metabolic regulation of embryonic development.[66–68] In recent years, both *in vitro* and *in vivo* studies have contributed to our understanding of the mechanistic link between metabolic pathways and cell fate specification.[5,69] Here, we found an important role of glycolytic activity in the induction of endoderm and mesoderm using both glycolysis inhibitors and Glc-free medium. Most importantly, we demonstrate that glycolytic activity acts as a dose-dependent upstream regulator of two instructive signals, Nodal and Wnt signaling.

Differences in our findings compared with a prior study,[23] which suggested that cells differentiating into endoderm and mesoderm require a shift from high glycolysis to high OxPhos, might be attributed to the use of directed differentiation protocols involving growth-factor-containing germ-layer-specific media. In the mouse gastruloid system, cells of all three germ layers co-emerge without the need for exogenous signaling agonists. This allowed us to identify the importance of glycolytic activity in the induction of mesoderm and endoderm via Wnt signaling and Nodal signaling activation. Because agonists of these signaling pathways are part of the directed differentiation protocol, it is conceivable that the importance of glycolytic activity for the activation of these pathways could not be detected previously.

Our findings align with another recent study that showed adverse effects of 2-DG treatment on mesoderm specification in gastruloids, highlighting the reproducibility of this phenotype.[70] Contrary to our results, however, the authors did not observe a phenotype in Glc-free medium and concluded that glycolytic activity is dispensable for mesoderm development in gastruloids. A possible explanation might be that the experiments were conducted in the presence of CHIR, a condition that is analogous to our rescue experiments. Because even low amounts of Glc are sufficient for *T/Bra* induction, residual Glc after washing out high-Glc differentiation media might be another potential explanation for the observed differences.

In accordance with our findings, an important role of glycolysis has been suggested for mesoderm development at later stages of embryonic development, when bipotent NMPs commit to either a neural tube or PSM fate.[14] In NMPs it is, however, still disputed whether glycolytic activity has a promoting or inhibitory function on Wnt and Fgf signaling.[13,15,17] Concurrently with our research, another paper in this issue of *Cell Stem Cell* has identified metabolic activity as a key driver of phenotypic variation through integrated molecular-phenotypic profiling of trunk-like structures, a gastruloid-based *in vitro* model system for neural tube and somite formation.[71] Specifically, they demonstrate that an early imbalance in OxPhos and glycolysis leads to aberrant morphology and biases cells toward the neural lineage, consistent with our findings during germ layer formation.

Although the relationship between glycolysis and posterior NMP-derived mesoderm has been explored in several studies, little is known about the metabolic control of endoderm specification. Previous studies have found that glycolysis can promote endoderm differentiation.[72,73] One suggested mechanism involves the Lin41 protein kinase as a non-canonical phosphorylation target of glycolytic enzyme Pfkp, resulting in the suppression of *Sox2* and leading to increased endodermal differentiation of ESCs.[72] Here, we show that glycolytic activity is needed for the activation of endoderm-promoting Nodal signaling and, therefore, our data introduce an additional layer of metabolic control to our understanding of endoderm specification.

Bioenergetic vs. signaling function of glycolysis

Metabolism plays a vital role in generating the necessary building blocks and providing an energetically favorable environment for growth and proliferation. Differentiating cells might display unique bioenergetic demands that must be met to ensure the proper development of certain cell types.[74,75] For instance, the energetic requirements of prospective mesodermal and endodermal cells migrating through the PS might differ significantly from epithelial cells fated to become neuroectoderm. However, our finding that the phenotype caused by glycolysis inhibition can be rescued through the activation of Nodal and Wnt signaling pathways in the absence of glycolytic activity indicates that cell fate specification is not impaired due to energetic constraints. This suggests an important role of glycolytic activity as a positive regulator of developmental signaling pathways. The effect of glycolysis inhibition on gastruloid growth, however, clearly highlights the importance of the anabolic function of Glc metabolism during development. It is important to point out

that morphogenetic movements in the gastruloid model system are not comparable with *in vivo* gastrulation. Therefore, it is conceivable that glycolytic activity plays an additional role in the migration of cells *in vivo*, as previously suggested for neural crest and mesodermal cells.[8,16] Because the BMP4-induced micropatterned hESCs colonies undergo epithelial-to-mesenchymal transition followed by cell migration,[76] this model system offers an ideal platform for future studies on the impact of glycolytic activity on mesoderm and endoderm migration.

An open question is how changes in glycolytic activity can regulate Nodal, Wnt, and Fgf signaling during germ layer induction. Metabolite-driven post-translational modifications, moonlighting glycolytic enzymes, and metabolite-protein interactions are possible links between metabolism and signaling.[3,5,7] For instance, high glycolytic activity promotes acetyl-coenzyme A (CoA) levels that can be rate-limiting for protein acetylation.[77] Interestingly, the activity of Nodal and Wnt signaling transducers is known to be modulated by acetylation.[78–80] It has been also suggested that glycolysis-driven changes in the intracellular pH may further favor β-catenin acetylation and nuclear translocation during NMP differentiation.[15] Glycolytic activity also feeds into the production of the building blocks necessary for glycosylation.[81,82] Recently, it was proposed that 2-DG treatment affects mesoderm specification due to reduced glycosylation, based on the knowledge that several proteins involved in Wnt and Fgf signaling transduction are known to be glycosylated.[16,70,83]

Besides post-translational modifications, glycolytic activity has also been shown to affect cellular localization of glycolytic enzymes, thereby allowing them to perform non-canonical functions. For instance, translocation of Pfkl and Aldoa into the nucleus has been hypothesized to modulate Wnt signaling during somitogenesis.[13] Moreover, metabolites might also modulate protein activity by direct binding. Sentinel metabolites, such as fructose-1,6-bisphosphate (FBP), whose concentration changes with glycolytic activity, might be interesting targets for metabolite-protein allosteromes, as suggested by Miyazawa and colleagues.[13]

Most of the previous work has been focused on Wnt signaling and mesoderm specification. Currently, we know only little about how glycolytic activity modulates the Nodal signaling pathway and endoderm differentiation. Future work will be important to identify actual changes in acetylation and glycosylation patterns of signaling components and functionally link them to their activity, as well as to probe non-canonical glycolytic enzyme function and possible metabolite-protein interactions.

Relevance for *in vivo* mouse gastrulation and human development

The observation that exogenous Glc concentrations ranging from 0.02 to 2.5 mM have a dose-dependent effect on germ layer proportions raises the question of whether Glc levels within the embryonic environment fall within this regulatory range and whether glycolytic activity may function as an instructive cue during *in vivo* gastrulation. Measurements of Glc levels in uterine fluid suggest a concentration of approximately 0.6 mM.[30] However, the actual concentration the embryo experiences might be even lower, as Glc must be absorbed and transported through several extraembryonic epithelial layers to reach epiblast cells at the onset of gastrulation.[84–86] Because we found an almost linear dose dependency between Glc levels and T/Bra::GFP-expressing cells until 0.5 mM Glc (Figure S5E), it is conceivable that differential expression of Glc transporters and glycolytic enzymes could account for spatial variations in glycolytic activity, thereby impacting cell fate decision-making. This notion gains added significance when considered alongside recent findings that demonstrate the spatiotemporal coordination of Glc transporter expression during mouse gastrula development.[16] In the same study, the authors describe a wave of Glc metabolism activity with functional importance for PS formation, further corroborating our *in vitro* findings.

The validation of the effect of 2-DG on T/BRA expression in a human model for gastrulation, and its rescue by Nodal and Wnt agonists, suggests that the role of glycolytic activity as a positive regulator of signaling is conserved in human cells.

Our study provides new insights into the interplay between metabolism and signaling pathways that coordinate the differentiation of pluripotent cells into the three germ layers. The demonstrated role of exogenous Glc concentration on cell fate decision-making represents an initial stride toward establishing nutritional control of cell type composition in complex *in vitro* model systems. Future work in this emerging research field will further improve our understanding of how metabolism is integrated into cellular behavior and how metabolic conditions affect embryonic development.

Limitations of the study

Our study provides evidence for a crucial role of glycolytic activity in the induction of endodermal and mesodermal cell lineages. These results are based on *in vitro* model systems of germ layer differentiation for both mouse and human development. It is noteworthy that although the two stem cell-based models of development (mouse gastruloids and micropatterned human gastruloids) are useful to overcome challenges of accessibility and nutritional manipulation of the *in vivo* embryo, they recapitulate limited aspects of early mammalian gastrulation. Future studies that can test these findings with the *in vivo* setting will be important.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to, and will be fulfilled by, the lead contact, Vikas Trivedi (vikas.trivedi@embl.es).

Materials availability

Plasmids and cell lines generated in this study are available upon request.

Data and code availability

RNA-seq data have been deposited at BioStudies (BioStudies: E-MTAB-13543) and GEO (GEO: GSE282050). Original code has been deposited: https://git.embl.de/grp-mif/image-analysis/gastr-volume-hcr. Microscopy data reported in this paper will be shared by the lead contact upon request. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization: K.S.S., M.E., and V.T.; methodology: K.S.S., M.E., and V.T.; software: N.G.; validation: K.S.S., E.H., L.S.P., L.Y., K. Arató, and I.H.; formal analysis: K.S.S., P.P., L.S.P., K.W., K. Anlas, M.O.L., K.R.P., and I.H.; investigation: K.S.S., E.H., L.S.P., K. Arató, and L.Y.; writing – original draft: K.S.S. and V.T.; visualization: K.S.S. and V.T.; supervision: K.S.S., M.E., and V.T.; project administration: V.T.; funding acquisition: K.S.S., I.H., M.E., and V.T.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
Ndiff	Takara	Cat# 40002
2-Deoxyglucose	Sigma	Cat# D8375
NaN_3	Sigma	Cat# S8032
Activin-A (mouse cells)	R&D Systems	Cat# 338-AC
Activin-A (human cells)	R&D Systems	Cat# AFL338
CHIR99021 (mouse cells)	Merck	Cat# SML1046
CHIR99021 (human cells)	Tocris	Cat# 4423
BMP4	R&D Systems	Cat# 314BP/CF
Fgf8	Tebubio	Cat# AF-100-25
Glucose	Agilent	Cat# 103577-100
Critical commercial assays		
HCR™ RNA-FISH Bundle	Molecular Instruments	https://store.molecularinstruments.com/
Seahorse XF Real-Time ATP Rate Assay Kit	Agilent	Cat# 103592-100
L-Lactate Assay Kit	abcam	Cat# ab65330
NAD/NADH-Glo Assay	Promega	Cat# G9071
Deposited data		
Single gastruloid RNA sequencing data	This paper	BioStudies: E-MTAB-13543
Single-cell RNA sequencing data	This paper	GEO: GSE282050
Experimental models: Cell lines		
E14 mESCs T/Bra::GFP	Laboratory of Alfonso Martinez Arias	originating from Fehling et al.[87]
E14 mESCs T/Bra::GFP, CAG::SUC2-CD8HA, PGK::mcherry	This paper	N/A
H9 hESCs	WiCell	hPSCReg #WAe009-A
ESI07 hESCs	ESIBIO	hPSCReg #SSCCe003-A
Oligonucleotides		
HCR probe: *Aldh1a2*	Molecular Instruments	Acc# NM_009022
HCR probe: *Eomes*	Molecular Instruments	Acc# NM_010136.3
HCR probe: *Gata6*	Molecular Instruments	Acc# NM_010258
HCR probe: *Meox1*	Molecular Instruments	Acc# NM_010791
HCR probe: *Nanog*	Molecular Instruments	Acc# NM_001289828.1
HCR probe: *Sox1*	Molecular Instruments	Acc# NM_009233.3
HCR probe: *Sox17*	Molecular Instruments	Acc# NM_011441.5
HCR probe: *Sox2*	Molecular Instruments	Acc# NM_011443.4
HCR probe: *T/Bra*	Molecular Instruments	Acc# NM_009309.2
Antibodies		
rabbit anti-SOX2 (1:200)	Cell Signaling Technology	Cat# 3579S; RRID: AB_2864411
goat anti-TBXT/BRA (1:300)	R&D Systems	Cat# AF2085; RRID: AB_2200235
mouse anti-Smad2/3 (1:100)	BD Biosciences	Cat# 610843; RRID: AB_398162
donkey anti-mouse IgG, Alexa Fluor 488 (1:500)	Thermo Fisher Scientific	Cat# A21202; RRID: AB_141607
donkey anti-rabbit IgG, Alexa Fluor 555 (1:500)	Thermo Fisher Scientific	Cat# A31572; RRID: AB_162543
donkey anti-goat IgG, Alexa Fluor 647 (1:500)	Thermo Fisher Scientific	Cat# A21447

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
MOrgAna v.0.1	https://github.com/LabTrivedi/MOrgAna/	Gritti et al.[88]
FIJI/Image J	NIH	https://imagej.net/Fiji
Python	Python Software Foundation	https://www.python.org
ilastik v.1.4	https://www.ilastik.org/	Berg et al.[89]
Python script for analysis of gene expression areas	This paper	https://git.embl.de/grp-mif/image-analysis/gastr-volume-hcr
FlowJo™	BD Life Sciences	https://www.flowjo.com
Wave online app	Agilent	https://www.agilent.com/en/product/cell-analysis/real-time-cell-metabolic-analysis/xf-software/agilent-seahorse-analytics-787485
Graphpad Prism	Graphpad Software	https://www.graphpad.com/
Phobius	Signal peptide prediction	https://phobius.sbc.su.se/
Other		
96-well ULA plates	Greiner	Cat#650970
Flat-bottom CellCarrier 96 ultra plate	PerkinElmer	Cat#6007008
MicroClime environmental microplate lid	Labcyte	Cat#LLS-0310
Seahorse cell culture plates	Agilent	Cat# 100777-004

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Mouse embryonic stem cell culture

E14 T/Bra::GFP mESCs[87] (kindly shared by the Martinez Arias lab) were cultured as previously described.[28] T/Bra::GFP mESCs were maintained in ESL medium on 0.1% gelatin-coated (Millipore, ES-006-B) tissue culture-treated 25cm^2 flasks (Corning, #CLS430639) in a humidified incubator (37°C, 5% CO2). ESL (KnockOut DMEM (Life technologies, 10829-018) supplemented with 10% ES tested fetal bovine serum (Life technologies, 16141-079), 1x Non-essential amino acids (Life technologies, 11140-035), 50 U/ml Pen/Strep (Life technologies, 15140-122), 1x GlutaMax (Life technologies, 35050-0610), 50μM 2-Mercaptoethanol (Life technologies, 31350-010) and 1000U/ml leukemia inhibitory factor (Life technologies, PMC9484)) was prepared and provided by the Tissue Engineering Unit of the Centre for Genomic Regulation. The Tissue Engineering Unit performs extensive batch testing for the fetal bovine serum and leukaemia inhibitory factor. Cells were passaged every 2-3 days and the ESL was exchanged on alternating days. Cells were cultured up to 50-80% confluency prior to passaging or experimental use. Mycoplasma testing was performed regularly.

Human embryonic stem cell culture

The cell lines used were the embryonic stem cell line H9 (XX, WiCell, WAe009-A), and ESI017 (XX, Esibio, SSCCe003-A). The identity of these cells as pluripotent stem cells was confirmed by staining of pluripotency markers OCT3/4, SOX2, NANOG. All cells were routinely tested for mycoplasma contamination, and negative results were recorded. Pluripotent stem cells were cultured in the chemically defined mTeSR1 media (StemCell Technologies, 85850) on Cultrex (R&D Systems)-coated tissue culture plates. Whole-colony routine passaging was done using L7,[90] and single-cell suspension for seeding experiments was generated using Accutase.

METHOD DETAILS

Mouse gastruloid generation and culture

Gastruloids were generated as previously described[28] with following modifications: In none of the experiments, we have treated gastruloids with CHIR between 48-72hpa as indicated in the original protocol. While the CHIR treatment is beneficial for gastruloid elongation, robust symmetry breaking and elongation is achieved in the absence of CHIR addition when E14 T/Bra::GFP mESCs were maintained in ESL.

Inhibitor and signalling agonist treatments in mouse gastruloids

Gastruloids were made by aggregating 300 cells in 40μl N2B27 (Ndiff227, Takara, Y40002). At 24hpa selected inhibitors and signalling agonists were applied for 24h (till 48hpa) as 40μl 2x solutions in N2B27. Following final concentrations were used: glycolysis inhibition with 6mM 2-DG (stock solution: 2.5M 2-DG in ddH_2O, Sigma, D8375), OxPhos inhibition with 1mM NaN_3 (stock solution: 1M NaN_3 in ddH_2O, Sigma, S8032), Nodal signalling activation with 25ng/ml Act-A (stock solution: 100μg/ml Act-A in 2mM HCl, R&D Systems, 338-AC-010), Wnt signalling activation with 3μM CHIR (stock solution: 10mM CHIR in DMSO, Sigma, #SML1046), Fgf

signalling activation with 100ng/ml Fgf8 (Tebubio, AF-100-25). At 48hpa, inhibitors and signalling agonists were washed out by adding 150μl of fresh N2B27, removing 180μl medium and adding another 150μl of fresh N2B27. At 72hpa, 140μl medium was removed and 150μl of fresh N2B27 medium was added.

Glucose concentration and washout experiments in mouse gastruloids

For Glucose (Glc) titration and washout experiments, homemade N2B27 was prepared by mixing equal parts of SILAC DMEM/F12 (Gibco, A2494301) and Neurobasal medium without glucose or pyruvate (Gibco, A2477501) and adding 0.5x N2 supplement (Gibco, 17502001), 0.5x B27 supplement (Gibco, 17504001), 1x GlutaMAX (Gibco, 35050061), 1mM sodium pyruvate (Gibco, 11360-070), 1x MEM non-essential amino acids (Gibco, 11140-050), 0.1mM 2-mercaptoethanol (Gibco, 31350-010) and respective concentrations of Glc (Agilent, 103577-100).

For Glc concentration experiments, 700 cells (if not stated otherwise) were aggregated in 40μl N2B27 with respective Glc concentration. At 24hpa, further 40μl N2B27 with respective Glc concentration were added. At 48hpa, further 110μl N2B27 with respective Glc concentration were added.

For Glc washout experiments, 350 cells were aggregated in 40μl N2B27 with 0.5mM Glc. At 24hpa, Glc was washed out by repeated washes with Glc-free medium (+170μl, -160μl, +160μl, -160μl, +160μl, -160μl Glc-free N2B27 medium). The washes were performed in the 96-well plate and the estimated Glc concentration in each well after the washes was <0.004mM. At 48hpa, 150μl of N2B27 medium with 0.5mM Glc was added.

HCR staining and imaging of mouse gastruloids

In situ mRNA detection was performed with the HCR™ RNA-FISH Technology (Molecular Instruments). Gastruloids were fixed for 40min in 4% paraformaldehyde (ChemCruz, SC-281692) and subsequently washed 3 times in PBS without Mg_2/Cl_2 (PBS-/-, Sigma, D8537). Gastruloids were then dehydrated in MeOH and subsequently stored in fresh MeOH at -20°C for overnight or longer. Gastruloids were rehydrated by sequential washes in 75% MeOH, 50% MeOH and 25% MeOH in PBS-/- supplemented with 0.1% Tween20 (Sigma, P9416) (PBST) and finally washed five more times in PBST. Gastruloids were prehybridised in probe hybridisation buffer (PHB, Molecular Instruments, HCR™ RNA-FISH Probe Set) for at least 30min at 37°C. Pre-PHB was then replaced with PHB containing 0.4pmol/100μl of each probe of interest (Molecular Instruments, commercially designed probes using accession# as indicated in key resources table) and samples were incubated at 37°C overnight. Samples were then washed four times in probe wash buffer (PWB, Molecular Instruments) for 15min each at 37°C and two times in 5x sodium chloride sodium citrate with 0.1% Tween (5x SSCT, prepared in-house) for 5min at RT, each. Next, gastruloids were pre-incubated in amplification buffer (Molecular Instruments) for 30min at RT. In the meantime, all h1 and h2 hairpins (Molecular Instruments) of one reaction were separately mixed, heated to 95°C for 90s and cooled for 30min in the dark at RT. Then, h1 and h2 hairpins of each reaction mix were combined in amplification buffer at RT to a final concentration of 60nM per hairpin. The amplification buffer containing the hairpins was added to the samples after removing the pre-amplification buffer and samples were incubated overnight in the dark at RT. On the last day, five washes in 5x SSCT were conducted at RT: 2x 5min, 2x 30min and 1x 5min. Samples were stored at 4°C in the dark for up to 2 weeks before imaging on an Opera Phenix HSC system (PerkinElmer).

Wide field imaging and time lapse microscopy of mouse gastruloids

For live imaging, ULA 96-well plates containing gastruloids were transferred into the Opera Phenix HSC system (Perkin Elmer) and snapshots taken every 24h at 37°C and 5% CO_2 using a 10x air (NA 0.3) objective in wide field mode. The same settings were applied for time lapse imaging of gastruloids in ULA 96-well plates covered with MicroClime environmental microplate lids (Labcyte, LLS-0310) filled up with sterile H_2O or PBS (Thermo Fisher Scientific) to avoid medium evaporation. Images were taken every 5min for several days with medium changes every 24h.

For imaging of fixed gastruloid samples on the Opera Phenix HSC system, stained gastruloids were transferred into individual wells of flat-bottom CellCarrier 96-well (Perkin Elmer, 6007008) in SSCT. Images were acquired in confocal mode using a 20x water objective.

Generation of human micropatterned stem cell colonies

For micropatterned colonies, we followed Deglincerti et al.[91] In short, cells were seeded as a single-cell suspension onto laminin-coated micropatterns in mTeSR1 with ROCK inhibitor. Unbound cells were washed 35 minutes after seeding with PBS-/- and mTESR1 with ROCK inhibitor was swapped for mTESR1 without ROCK inhibitor after 2h. hPSC micropattern colonies were maintained for 24 hours after seeding before differentiation. After 24h differentiation was initiated by changing media for mTeSR1 without ROCK inhibitor and with 50 ng/ml BMP4 (R&D Systems, #314BP/CF). BMP4 treatment was maintained throughout the experiment. Between 24-48h after differentiation initiation (hpdi) with BMP4, cells were treated with glycolysis inhibitor and signalling activators as indicated in the figures. Following final concentrations were used: glycolysis inhibition with 5mM 2-DG (Sigma, D8375), Nodal signalling activation with 50ng/ml Act-A (R&D Systems, #AFL338), Wnt signalling activation with 3μM CHIR (Tocris, #4423). All experiments were done in micropatterned 18-well Ibidi slides made using the protocol described in Freeburne et al.[92] All colonies were 700 μm in diameter. Reagents to modify signalling (listed in key resources table) during pattern formation were added at the times indicated in the figures.

Immunostaining of human micropatterned stem cell colonies

Micropatterned colonies in 18-well Ibidi slides were rinsed with PBS-/-, fixed for 20 min in 4% paraformaldehyde, rinsed twice with PBS-/-, and blocked for 30 min at room temperature with 3% donkey serum and 0.1% Triton X-100 in 1× PBS-/-. After blocking, cells were incubated with primary antibodies at 4°C overnight, followed by three washes in PBST (PBS-/- with 0.1% Tween 20). They were then incubated with secondary antibodies and DAPI for 30 min at room temperature and washed twice in PBST at room temperature. Antibodies and working dilutions can be found in the key resources table.

Imaging of human micropatterned stem cell colonies

Imaging was done on an Andor Dragonfly/Leica DMI8 spinning disk confocal microscope with a 40x, NA 1.1 water objective. Segmentation and quantification protocols can be found in Jo et al.[93]

Bulk RNA-seq sample preparation

We performed bulk RNA-seq on single gastruloids (2-DG treated for 24h and control, n=6 each) at 48hpa (0h after 2-DG treatment) and at 96hpa (48h after 2-DG treatment). RNA was extracted from fresh single gastruloids using the RNeasy Micro Kit (QIAGEN). RNA quality control was performed with the Agilent RNA 6000 Pico Kit (Agilent, 5067-1513) on the Bioanalyzer system (Agilent).

Bulk RNA-seq library preparation and sequencing

The extracted RNA was concentrated using a vacuum concentrator centrifuge (SpeedVac, Eppendorf, EP5305000100-1EA) to a final volume of 3μl. Then 1μl of 10mM dNTPs (Kappa, KK1017) and 1μl of oligo-dT primer (5′-AAGCAGTGGTATCAACGCAGAG TACT30VN-3, custom from IDT) were added to the samples. Then full-length cDNA sequencing libraries were prepared following a modified version of the SMART-Seq2 protocol using SuperScript IV RT (SSRT, ThermoFisher 18090050) and the tagmentation procedure previously described.[94,95] Briefly, a reverse transcription reaction was performed with the following mixture per sample and cycling conditions: 2μl SSRT IV 5x buffer (ThermoFisher 18090050), 0.5μl 100mM DTT (ThermoFisher 18090050), 2μl 5M betaine (Merck, B0300-5VL), 0.1μl 1M $MgCl_2$ (Ambion, AM9530G), 0.25μl 40 U/μl RNAse inhibitor (Clontech Takara, 2313A), 0.25μl SSRT IV (ThermoFisher 18090050), 0.1μl 100μM TSO primer (5′-AAGCAGTGGTATCAACGCAGAGTACATrGrG+G-3′) and 1.15μl RNase-free H_2O (Ambion, AM9916) at 52°C for 15min, followed by an incubation at 80°C 10min. The resulting cDNA was amplified adding 12.5μl of KAPA HiFI HotStart Ready Mix 2x (Roche, 07958935001), ISPCR primer at 5μM (5′-AAGCAGTGGTATCAACGCAGAGT-3′) and 2.30μl of H_2O. The cycling conditions were 98°C for 3min, followed by 18 cycles of 98°C for 20s, 67°C for 15s and 72°C for 6min and a final incubation at 72°C for 5min. After amplification, individual libraries were size selected using magnetic SPRI bead purification (Beckman coultier, B23319) (0.6x) omitting the ethanol wash steps and using an elution volume of 13μl H_2O. For the next steps, the clean cDNA was normalized to 0.2ng/μl. In short, 1.25μl of each sample was taken into a tagmentation reaction containing 1.25μl of Dimethylformamide (Merck, 227056), 1.25μl of tagmentation buffer (40mM Tris-HCl pH 7.5, 40mM MgCl2) and 1.25μl of an in-house generated and purified Tn5 (previously prepared as described by Hennig et al.[94] and diluted to a 1:300 ratio with water). The mixture was incubated at 55°C for 3min. After that, 1.25μl of 0.2% SDS (Merck, 71736) was added to stop the tagmentation reaction followed by 5min incubation at room temperature. The resulting fragments were then amplified by PCR using 6.75μl of KAPA 2x HiFi master mix, 0.75μl of Dimethyl sulfoxide (Merck, D8418-50ML) and 2.5μl of dual indexed primers. Cycling conditions were 72°C for 3min and 95°C for 30s, followed by 12 cycles of 98°C for 20s, 58°C for 15s, 72°C for 20s and a final incubation at 72°C for 3min. Final libraries were size selected with the magnetic SPRI beads (0.8x) and were quantified using Qubit HS assay (Thermo Fisher Scientific, Q33231). A Bioanalyzer system (Agilent, 5067-4626) was used to assess library sizes and fragment distribution. Finally, libraries were pooled equimolarly and the final pool underwent an extra round of size selection using magnetic SPRI beads at a 0.7x ratio. Samples were single end sequenced as a 24-plex on Illumina NextSeq2000 using a P2 100 cycles kit (Illumina, 20046811).

scRNA-seq sample preparation

Respective number of gastruloids of the four different conditions at 96hpa were collected from 96-well plates and transferred to a 15ml falcon (control: n=50; 2-DG, 2-DG+ActA, 2-DG+CHIR: n=75). Medium was removed and gastruloids were washed with pre-warmed PBS -/- (Sigma, D8537). PBS-/- was removed and 1ml TrypLE (Thermo Fisher Scientific, 12604) was added. Samples were incubated for 5min at 37°C with repeated flicking. Cell suspension was then gently pipetted up and down to ensure a single-cell suspension. Subsequently, 4ml of ice-cold 0.1% RNAse-free BSA in PBS-/- (BPBS) was added and cell suspension was centrifuged at 170g for 3min at 4°C. All following steps were performed on ice/at 4°C. Cells were washed 2 times in BPBS, before they were resuspended in 300μl BPBS using a wide bore P1000 tip. Cell suspension was then passed through a cell strainer (Corning, 352235), counted and diluted to a target concentration of 1.0-1.5x10^6 cells/ml.

scRNA-seq library preparation and sequencing

Library preparation and sequencing was performed by the CRG Genomics Unit. Cells were barcoded using a Chromium GEM-X Single Cell 3' Reagent Kits v4 (10x Genomics) according to manufacturer's instructions. Libraries were sequenced with a NovaSeq 6000 system (Illumina).

Seahorse metabolic rate assay

To mimic a differentiation timeline similar to gastruloid development, mESCs were seeded at a density of 1.67x10^5 cells/T25 flask in ESL medium. The following day cells were washed twice in PBS (Sigma, D8662) before adding N2B27. After 24 hours, medium was replaced with N2B27 supplemented with metabolic inhibitors and/or signalling agonists, as indicated for the different experiments. After 24h of treatment, cells were dissociated with trypsin-EDTA (Thermo Fisher Scientific) for 1 min at 37°C, harvested in their respective culture media, washed and resuspended in N2B27 medium +/- inhibitors/signalling agonists. 80,000cells/well in 100µl medium were seeded into fibronectin-coated Seahorse cell culture plates (Agilent, 100777-004) and left at RT for 15 min to ensure equal distribution of cells within all microwells. Plates were then transferred into a 37°C incubator for 2h for complete attachment. Afterwards, cells were washed carefully in Seahorse XF DMEM medium (Agilent, 103575-100) supplemented with 1mM pyruvate (Agilent, 103578-100), 2mM glutamine (Agilent, 103579-100) and the respective concentration of Glc (Agilent, 103577-100) +/- inhibitors/signalling agonists. If not indicated otherwise, 10mM Glc were used as a standard concentration. During the washing steps, 950µl Seahorse medium were added to the attached cells, 1000µl removed, another 450µl added and removed and a final 450µl added onto the cells (final volume 500µl). 500µl Seahorse medium was added to each of the 4 background wells. Cells were then incubated at 37°C without CO_2 for 45 min. The Seahorse cartridge was hydrated overnight and loaded with components of the real-time ATP rate assay (Agilent, 103592-100) as indicated in the manual (final inhibitor concentration: 1.5µM oligomycin, 0.5µM rotenone and 0.5µM antimycin A). All samples were run with at least 4 technical replicates in a Seahorse XFe24 Analyzer (Agilent).

Extracellular lactate assay

To measure cellular lactate production in different Glc concentrations, mESCs were seeded into 48-well plates at a density of $5x10^4$ in ESL medium. The following day cells were washed twice in PBS before adding 1ml N2B27. After 24h, cells were washed again twice and medium was replaced with 0.5ml homemade N2B27 supplemented with 0mM, 0.02mM, 0.1mM, 0.5mM, 2.5mM or 12.5mM Glc. After 1h medium was replaced with another 0.5ml homemade N2B27 with respective Glc. Supernatant was collected for measurement after 1h of incubation. After centrifugation of the supernatant, samples were kept on ice for immediate use or frozen at -80°C.

To measure the ability of different metabolites to restore glycolytic activity/lactate production in the absence of glucose in the medium, supernatant was collected from gastruloid cultures at 48hpa, 24h after the removal of glucose/addition of rescue metabolites. After centrifugation of the supernatant, samples were kept on ice for immediate use or frozen at -80°C.

Fluorimetric lactate measurements were performed using the Lactate Assay Kit (abcam, ab65330) following manufacturer's instructions with some modifications: the reaction volume was reduced to 50µl and 1-25µl of 1:10 diluted samples were used for analysis to obtain raw values within the range of a lactate standard curve. Fluorescence was measured using a Tecan Infinite M Plex plate reader and an Excitation/Emission of 523nm/587nm, respectively.

NAD^+/NADH assay

To measure intracellular NAD^+/NADH ratio in cells cultured in different Glc concentrations, mESCs were seeded into 48-well plates at a density of $5x10^4$ in ESL medium. The following day cells were washed twice in PBS before adding 1ml N2B27. After 24h, cells were washed again twice and medium was replaced with 0.5ml homemade N2B27 supplemented with 0.02mM, 0.1mM, 0.5mM, 2.5mM or 12.5mM Glc. After 1h medium was replaced with another 0.5ml homemade N2B27 with respective Glc concentrations for 1h.

Cells were then washed twice in PBS and lysed in 350µl 0.5% DTAB in 0.1N NaOH and 50µl replicates were transferred into a white flat-bottom microwell plate (Thermo Fisher, 611F96WT). One half of the samples was treated with 25µl 0.4N HCl (NAD^+ samples) before the entire plate was incubated at 60°C for 15min and then left at RT for 10min. NAD^+ samples were neutralized with another 25µl 0.5M Trizma base and residual samples treated with 50µl 1M Trizma-HCl (NADH samples). NAD^+/NADH-Glo Detection Reagent (Promega, G9071) was prepared and measurements performed according to the manufacturer's instructions. A final ratio of intracellular NAD^+/NADH was calculated for each Glc concentration.

Flow cytometry

To determine how size and Glc concentration affect the percentage of T/Bra positive cells, gastruloids from 4 different initial cell numbers (350, 700, 1050 and 1400 cells) were grown at 0.02mM, 0.1mM, 0.5mM and 2.5mM Glc until 48hpa. 24 gastruloids of each condition (48 for initial cell number of 350 cells) were pooled and washed in PBS-/- before adding 50µl of accutase (Gibco, A1110501). After 5 min of incubation in accutase, gastruloids were dissociated by adding flow buffer (20% N2B27 with 0.02mM Glc and 2.5mM EDTA in PBS-/-) and gentle up and down pipetting. Cells were then pelleted at 170xg for 3min and resuspended in 200µl flow buffer. Flow cytometry was performed on BD LSRFortessa Cell Analyzer (BD Biosciences).

Generation of SUC2-CD8 expressing mESCs line

To generate mESCs that can utilise sucrose, we generated a membrane-tethered version of the beta-fructofuranosidase SUC2 (SGD: S000001424) from *Saccharomyces cerevisiae S288C*. We first designed the coding sequence for such a protein *in silico*. To ensure efficient secretion, we identified the yeast signal peptide using the signal peptide prediction tool Phobius (https://phobius.sbc.su.se/) and exchanged it with the signal peptide of the interleukin-2 receptor alpha chain (ENSMUSG00000026770) from mouse. Membrane-tethering was achieved by the C-terminal fusion of HA-tagged CD8 (ENSMUSG00000053977). To ensure that SUC2 activity is not affected by the fusion to CD8, we included the sequence for a flexible linker (GSGSGSGSGGGSGGG) between the two proteins.

The SUC2-CD8 coding sequencing was codon-optimised for mouse and synthesised using the GeneArt Gene Synthesis service (Thermo Fisher Scientific).

Next, the SUC2-CD8 coding sequence was cloned under the control of a CAG promoter into a vector that also contained an mCherry coding sequence under a PGK promoter. The construct was then stably transfected using PiggyBac system[96] and Lipofectamine Stem Transfection Reagent (Thermo Fisher Scientific) into the E14 T/Bra::GFP mESC line. Cells were then FACS sorted for mCherry positive cells on day 5 and then again on day 14 after transfection.

QUANTIFICATION AND STATISTICAL ANALYSIS

Gastruloid size and T/Bra intensities

Mean T/Bra::GFP intensity (with background subtraction) and gastruloid size at 48hpa were quantified using MOrgAna v.0.1.[88] In brief, after generation of segmentation masks using the brightfield channel of widefield images obtained with the Opera Phenix HSC system (Perkin Elmer), MOrgAna was used to calculate the area and the mean T/Bra::GFP intensity (with background subtraction).

Kymographs

Kymographs were generated by averaging several replicas (n numbers are indictaed in the figure legends) using MOrgAna v.0.1.[88] In brief, after generation of segmentation masks and calculation of AP fluorescence profiles for each gastruloid across the entire time-lapse, all experimental conditions to be compared were loaded simultaneously as separate groups of respective gastruloids. T/Bra::GFP fluorescence intensities along the thereby defined AP axis were normalized to the global maximum (yielding values between 0-1). The colour code denoting T/Bra fluorescence intensity was manually set between 0-0.83 (Figure 1B) and 0.05-0.95 (Figure 2D) normalized units to enable clear visual comparison of fluorescence intensities between conditions. Prior to normalization, background fluorescence as a result of the culture media or gastruloid tissue properties was removed either by subtracting the average fluorescence intensity of the non-mask area or by setting the minimum pixel intensity level found in the gastruloid mask to 0.

Analysis of area of gene expression

The germ layer proportions were measured as the relative area of gene expression from the images of HCR stained gastruloids and obtained using custom written software using Python. Individual field of views (tiles) obtained from imaging on the Opera Phenix microscrope were flat-field corrected using the parameters obtained from the Harmony software (version 4.9), and full-well views were generated stitching the tiles with the Grid/Collection Stitching plugin of FIJI.[97] Semantic segmentation was performed with a pixel classification network in the ilastik software v. 1.4[89] to classify pixels belonging to gastruloids or background using the bright field images. To perform instance segmentation, that is, identification of individual gastruloids in each well, we eroded the ilastik masks to separate individual gastruloids, removed objects smaller than $1000\mu m^2$ and computed a distance transform of the masks using the Python package scipy.[98] The locations of the max values of the distance transform image were used as seeds of a watershed segmentation using scikit-image.[99] Finally, using the histogram of pixel values from all gastruloid masks in the image, we obtained a threshold value for each fluorescent channel using the otsu algorithm, and computed gene expression masks. The relative gene expression area was computed as the total number of bright pixels in each gastruloid divided by the total area of the same gastruloid. To avoid bias in threshold computation under conditions in which no gene expression was found, in each experiment, we always had one control condition in which the otsu thresholds were computed, and subsequently the same thresholds were applied to all other metabolic conditions in the same experiment. When the otsu algorithm failed to generate the correct thresholds, we set such thresholds manually by visual inspection. The Python code used to perform image analysis is publicly available and deposited at: https://git.embl.de/grp-mif/image-analysis/gastr-volume-hcr.

Image analysis of human micropatterned stem cell colonies

Segmentation and quantification protocols can be found in Jo et al.[93]

Bulk RNA-seq analysis

We have aligned the sequencing reads to GRCm39 reference (version 107) using STAR Aligner, version: 2.7.9a[100]; Link: https://github.com/alexdobin/STAR. Aligned reads were counted using HTSeq-count, version 2.0.2[101,102]; Link: https://htseq.readthedocs.io/en/master/. Differential gene expression was assessed with *DESeq2*.[103] Log_2 fold changes between vehicle and 2-DG treated gastruloids were shrunk using rlog function. RNA-seq data have been deposited in the BioStudies, and are available under the BioStudies: E-MTAB-13543.

scRNA-seq quality control and data analysis

Fastq files were generated and demultiplexed from raw base call (bcl) files using cellranger mkfastq pipeline.[104] Alignment, filtering, barcode, and unique molecular identifier (UMI) counting were conducted with cellranger-8.0.1 count pipeline using the mouse reference assembly (mm10) and annotation available on 10x genomics website with default settings (https://www.10xgenomics.com). The files were loaded into R Studio (v.4.4.1) using the 'load10X' function from the Seurat 5.1.0 package.[105] Samples were then

merged to create a single Seurat object. We discarded genes not expressed in at least 3 cells, low-quality cells, and putative cell artifacts from this object. Cells with less than 2500 genes and 6000 UMIs detected, or less than 5% UMIs mapped to ribosomal proteins, were discarded. Cells with more than 15% UMIs mapped to mitochondrial transcripts were discarded. The final Seurat object contained 38435 high-quality cells (62.5% of cells kept) and 25390 genes for downstream analyses.

The dataset was normalized using the sctransform function regressing out the percent of mitochondrial and ribosomal UMIs as well as cell cycle score ("percent.mt", "percent.rRNA", "S.Score", "G2M.Score"). We selected 3000 variable genes to compute 30 Principal components (PCs). After manual inspection, we selected the first 15 PCs to calculate the UMAP plot and define cell clusters using the Louvain algorithm with multilevel refinement with a resolution of 0.5. Using default parameters, the Seurat function FindAllMarkers was used to find marker genes for each cluster. We used the functions DimPlot, VlnPlot, FeaturePlot, DotPlot and FeatureScatter to generate plots presented in the figures.[105] Cell proportions were calculated based on cell clustering, and ggplot2[106] was used to create stacked barplots.

Gastruloid scRNA-seq data was projected onto the embryo atlas (MouseGastrulationData package, all samples available) from Pijuan-Sala et al.[64] using scmap package[107] with standard parameters (500 features). Sankey diagram comparing two datasets was visualized using getSankey module of scmap package.

scRNA-seq data have been deposited under GEO: GSE282050.

Seahorse metabolic rate analysis

The Wave online app (Agilent) was used for analysis & Graphpad Prism (Graphpad) for plot generation.

Flow cytometry analysis

Data was analysed using the FlowJo software (BD Life Sciences).